

Graph Machine Learning in the Era of Generative AI

MLoG-GenAI Workshop — ACM SIGKDD Conference



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Résumé

This project implements a knowledge graph that models the themes of the MLOG-GenAI workshop, focusing on the evolving landscape of Graph Machine Learning (GML) in the era of Generative AI. The graph captures relationships between technologies, research challenges, application domains, and emerging opportunities introduced by foundation models. The resulting graph serves as both an educational visualization and an analytical tool for exploring the interaction between GML and Generative AI.

Introduction

Graphs are powerful structures used to model relationships in social networks, molecules, web pages, and knowledge systems. Graph Machine Learning (GML) focuses on learning patterns from such relational data.

Recent advances in Generative AI and large foundation models (LLMs) have pushed GML into a new era. These models provide new capabilities—such as automated knowledge graph construction—while introducing new challenges like integrating graph topology with text-based reasoning.

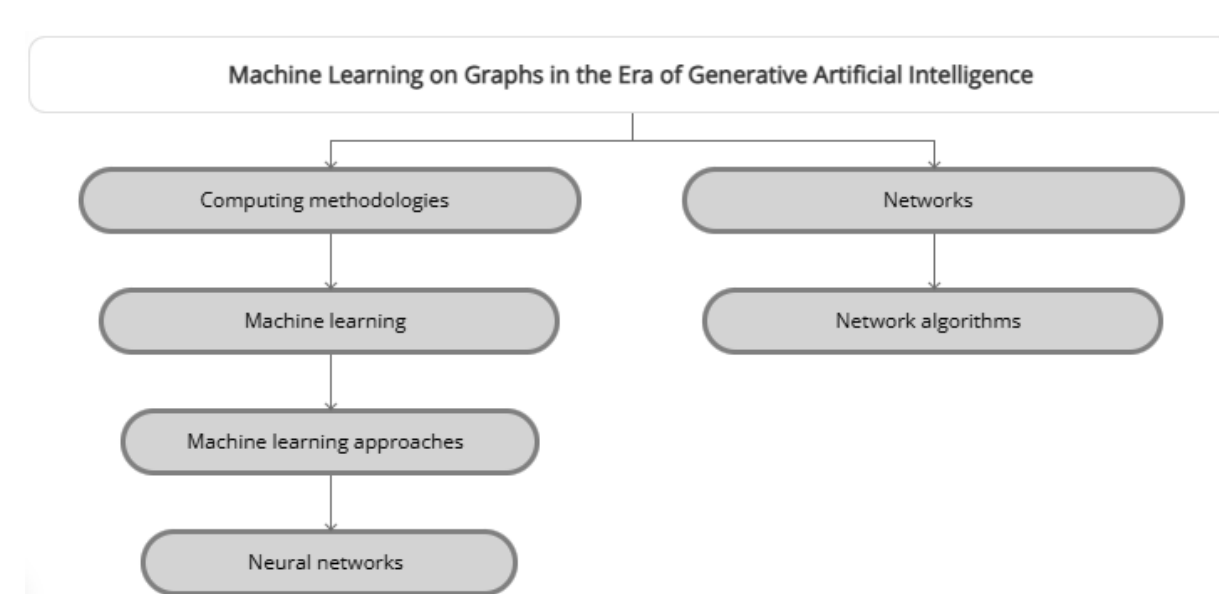


FIGURE 1 – ACM Computing Classification : Positioning MLoG-GenAI at the intersection of Networks and Machine Learning

Objectives

- Represent key GML concepts in a knowledge graph.
- Visualize connections between technologies, applications, and Gen-AI impacts.
- Provide an intuitive educational tool summarizing the MLoG-GenAI workshop.

Graph ML in the Gen-AI Era

The chosen theme explores how Graph Machine Learning evolves under the growing influence of Generative AI. The constructed knowledge graph highlights four major axes :

1. Evolution of GML Technologies

- **Shallow Embedding Models** — early node representation learning.
- **Graph Neural Networks** — They use message passing between nodes.
- **Graph Transformers** — They work like LLM transformers but on graph data.

- **Generative AI** — large foundation models reshaping graph reasoning.

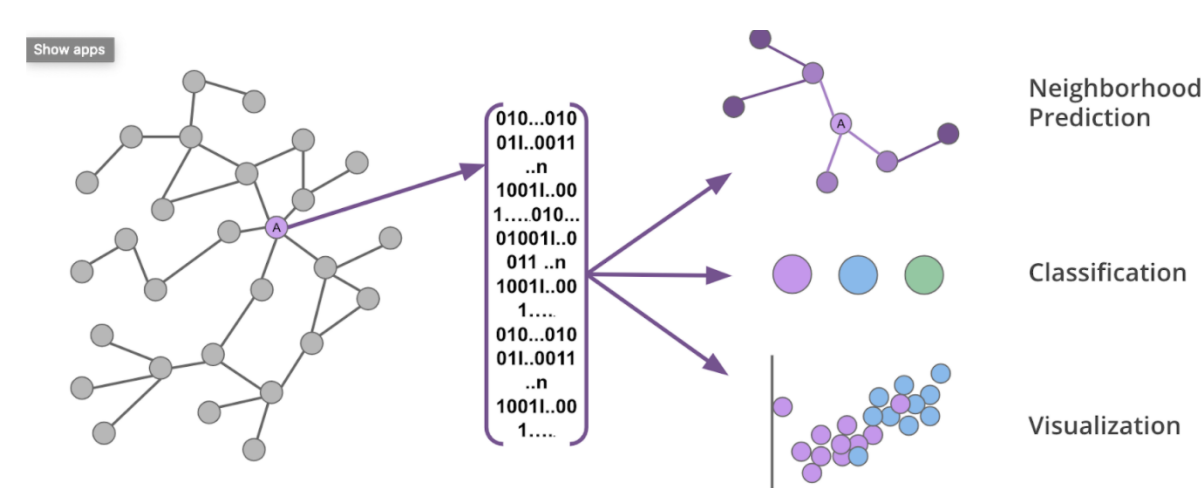


FIGURE 2 – ML applied on graphs

2. GML Application Domains

- Social networks .
- Drug and molecule discovery.
- Recommendation systems.
- Knowledge representation and reasoning.

3. Technical Challenges

- Too much information gets forced through too few edges.
- Scalability limits for large graphs.
- Difficulty injecting graph structure into LLMs.

4. Emerging Opportunities from Gen-AI

- Synthetic graph generation.
- LLM-assisted knowledge graph construction.
- Hybrid LLM-GNN architectures.
- Graph prompting for enhanced reasoning.

Data description

The data of this project is conceptual and originates from the workshop text. It is structured in Neo4j as nodes and relationships.

Node Types

- **Technologies** : Shallow Embeddings, GNNs, Graph Transformers, Gen-AI.
- **Mechanisms** : Message Passing, Attention, Positional Encodings.
- **Tasks** : Node Classification, Link Prediction, Graph Generation.
- **Challenges** : Oversmoothing, Scalability.
- **Opportunities** : Synthetic graph creation, reasoning automation.
- **Applications** : Social Networks, Drug Discovery, Recommender Systems.
- **Communities** : Academic researchers, industry practitioners.

- **Events** : MLoG-GenAI Workshop, ACM KDD Conference.

Relationships in the Graph

- **EVOLVED_TO** — technology progression.
- **APPLIED_TO** — use cases of GML.
- **USES** — mechanisms used by models.
- **IMPACTS / ENABLES** — Gen-AI influence.
- **FACES_CHALLENGE** — limitations faced by GML.
- **HAS_OPPORTUNITY** — new capabilities opened by Gen-AI.
- **HELD_AT** — workshop location.
- **BRINGS_TOGETHER** — research communities.



FIGURE 3 – Visualization of the knowledge graph.

Conclusion

This project demonstrates how scientific text can be transformed into a structured knowledge graph. The resulting graph provides an intuitive view of the evolution of GML, its applications, and the growing influence of Generative AI. It also highlights the challenges and opportunities emerging at this intersection, offering a foundation for deeper research and educational use.

Perspectives

Future work may include :

- Create an interactive dashboard to explore trends and relationships.
- Evaluate Gen-AI impact
- Use advanced NLP methods to build richer and more accurate knowledge graphs from scientific articles.

References

- Kipf, T. & Welling, M. (2017). *Semi-Supervised Classification with Graph Convolutional Networks*.
- Hamilton, W., Ying, R. & Leskovec, J. (2017). *Inductive Representation Learning on Large Graphs (GraphSAGE)*.
- ACM SIGKDD (2025). *MLOg-GenAI Workshop Description*.